Shor's Algorithm

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Here's the essence—clean and sharp—of what those pages are building toward.

⚙️ Classical vs Quantum (big picture)

- **Classical (Number Field Sieve)**
  Runtime ≈ subexponential → still *too slow* for large N
- **Quantum (Shor's Algorithm)**
  Turns factoring into a **period-finding problem** → efficiently solvable on a quantum computer

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🧠 Core Idea of Shor's Algorithm

You don't factor $N = pq$ directly.

You:

1. Pick a random a
2. Study the function $a^x \mod N$
3. Extract its **period** r
4. Use that period to recover factors

🔑 The Key Mathematical Trick

Once you find the period r:

$a^r \equiv 1 \pmod{N}$

This implies:

$$a^r - 1 \equiv 0 \pmod{N}$$

Factor it:

$a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1)$

Now the magic:

- One of these terms shares a **nontrivial factor with** N

So we compute:

$$p = \gcd(a^{r/2} - 1, N), \quad q = \gcd(a^{r/2} + 1, N)$$

→ Boom, factors found.

⚠️ Important Conditions

For this to work:

- r must be **even**
- $a^{r/2} \not\equiv -1 \pmod{N}$

If not → retry with different a

🔍 Why This Works (intuition)

You force:

- $a^r \equiv 1 \mod N$
- Which creates a **difference of squares**
- That splits into two numbers
- Each "leans" toward one of the hidden primes p or q

🧪 Example: $N = 15$

Pick $a = 2$

$$2^x \mod 15 \to \text{period } r = 4$$

Then:

$$a^{r/2} = 2^2 = 4$$

$$\gcd(4 - 1, 15) = 3, \quad \gcd(4 + 1, 15) = 5$$

✔️ Factors found: **3 and 5**

🚧 Bottleneck

- Hard part = **finding period** r
- Classical → slow
- Quantum → fast (via quantum Fourier transform)

💡 Final Insight

Shor doesn't "break factoring" directly.

It:

> **Transforms factoring into periodicity detection**
> → which quantum mechanics solves efficiently

If you want, I can next show:

- how the **quantum circuit actually finds** r
- or connect this to **RSA breaking step-by-step**

how to find period r

Short answer: **the whole game is finding** r for
$f(x) = a^x \bmod N$, where r is the smallest positive number such that
$a^r \equiv 1 \pmod{N}$

🧮 Classical way (what you can do by hand)

Just compute powers until the pattern repeats:

Example: $a = 2, N = 15$

$$2^1 \mod 15 = 2$$

$$2^2 \mod 15 = 4$$

$$2^3 \mod 15 = 8$$

$$2^4 \mod 15 = 16 \equiv 1 \mod 15$$

→ First time you hit **1** ⇒

$$r = 4$$

✔️ That's the period.

⚠️ Problem with classical method

- Works for small numbers
- For large N: **exponential time** ❌
  (you might need to try tons of powers)

⚡ Quantum way (Shor's real trick)

Instead of checking one-by-one, quantum computers:

Step 1: Superposition

Prepare all x at once:

$$|x\rangle \to |a^x \mod N\rangle$$

Step 2: Measure output

This collapses states into values spaced by multiples of r

Step 3: Apply QFT (Quantum Fourier Transform)

This extracts frequency information:

$$\text{You get something close to } \frac{k}{r}$$

Step 4: Classical post-processing

Use **continued fractions** to recover r

🧠 Intuition (important)

- Function $a^x \mod N$ is **periodic**
- Quantum computer "sees all repetitions at once"
- QFT converts repetition → **frequency**
- Frequency → gives you r

🔁 Summary

- **Definition:** smallest r such that $a^r \equiv 1 \mod N$
- **Classical:** brute force powers
- **Quantum:** detect periodicity using QFT (fast)

If you want, I can walk you through:

- a **step-by-step QFT example** (this is the real magic), or
- how continued fractions recover r from measurement output

Exercise 7.45. Use Shor's algorithm to factor $N = 35$.
(a) Pick a value of a such that $\gcd(a, N) = 1$ so that we can continue with the remaining steps of the algorithm.
(b) What is the period of $a^x \bmod N$? For the sake of this problem, just find the period classically since the numbers are small. Make sure the period r is even and $a^{r/2} \neq N - 1 \bmod N$. If not, go back and pick a different value for a.
(c) Calculate the factors $p = \gcd(a^{r/2} - 1, N)$ and $q = \gcd(a^{r/2} + 1, N)$.

solve a, b

Let's do it cleanly.

(a) Pick a

We need $\gcd(a, 35) = 1$.

Choose:

$$a = 2$$

✔ $\gcd(2, 35) = 1 \rightarrow$ valid

(b) Find the period r of $2^x \mod 35$

We look for the smallest r such that:

$2^r \equiv 1 \pmod{35}$

Compute:

- $2^1 \mod 35 = 2$
- $2^2 \mod 35 = 4$
- $2^3 \mod 35 = 8$
- $2^4 \mod 35 = 16$

- $2^5 \mod 35 = 32$
- $2^6 \mod 35 = 64 \equiv 29$
- $2^7 \mod 35 = 58 \equiv 23$
- $2^8 \mod 35 = 46 \equiv 11$
- $2^9 \mod 35 = 22$
- $2^{10} \mod 35 = 44 \equiv 9$
- $2^{11} \mod 35 = 18$
- $2^{12} \mod 35 = 36 \equiv 1$

✔️ First time we get 1 →

$$r = 12$$

Check conditions

- $r = 12$ is **even** ✔️
- $2^{r/2} = 2^6 = 64 \equiv 29 \not\equiv 34 \pmod{35}$ ✔️

(All good, no need to retry)

✅ Final Answers

- **(a)** $a = 2$
- **(b)** $r = 12$

If you want, I'll finish part (c) in one line—it gives you the factors instantly.

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part c

Now the payoff.

We already have:

- $a = 2$
- $r = 12 \Rightarrow r/2 = 6$

Step

$$2^6 = 64 \equiv 29 \pmod{35}$$

Compute factors

$$p = \gcd(29 - 1, 35) = \gcd(28, 35) = 7$$

$$q = \gcd(29 + 1, 35) = \gcd(30, 35) = 5$$

✅ Final Answer

$$p = 7, \quad q = 5$$

✔ So, $35 = 7 \times 5$

Exercise 7.46. Use Shor's algorithm to factor $N = 209$. Say we pick $a = 22$.
(a) Show that $\gcd(a, N) \neq 1$.
(b) What are the factors of N?

(a) Check $\gcd(a, N)$

Given $a = 22, \ N = 209$

$$\gcd(22, 209) = 11 \neq 1$$

✔ So condition fails — but this is actually *good*.

(b) Factors of N

When $\gcd(a, N) \neq 1$, Shor's algorithm **immediately gives a factor**:

$$p = 11$$

Then:

$$q = \frac{209}{11} = 19$$

✅ Final Answer

$$209 = 11 \times 19$$

💡 No need to find period here — you got lucky in Step 1 itself.